National Forest Cover Mapping Report

for the year 2023

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Acronyms

AD	Activity Data
AFOLU	Agriculture Forestry and Other Land Uses
BAU	Business as Usual
CEO	Collect Earth Online
COP	Conference of the Parties
CRGE	Climate Resilient Green Economy
EF	Emission Factor
EFCCC	Environment Forest and Climate Change Commission
EFD	Ethiopia Forest Development
FREL/FRL	Forest Reference Emission Level/ Forest Reference Level
GEE	Google Earth Engine
GFRA	Global Forest Resource Assessment
GHG	Greenhouse Gas
IPCC	Intergovernmental Panel on Climate Change
JRC	Joint Research Centre of Europe
LT-LEDS	Long Term Low Emissions Development Strategy
LULCC	Land Use Land Cover Change
LULUCF	Land use Land Use Change and Forestry
MEFCC	Ministry of Environment, Forest and Climate Change
MMU	Minimum Mapping Unit
$MtCO_2e$	Million tons of carbon dioxide equivalent
NDC	Nationally Determined Contribution
PA	Paris Agreement
RCMRD	Regional Centre for Mapping of Resources for Development
REDD+	Reducing Emissions from Deforestation and forest Degradation
SEPAL	System for Earth observation, data access, Processing and Analysis for Land monitoring
SOC	Soil Organic Carbon
SRTM	Shuttle Radar Topographic Mission
UNFCCC	United Nations Framework Convention on Climate Change

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1. Introduction

1.1 Background

Forest resources of Ethiopia make a significant contribution to the national economy. With forestry being primarily a rural activity, the sector has enormous potential to contribute to the transformation of the rural economy. The contribution by the sector should also be seen in the value that it adds through harboring biodiversity resources and other ecosystem services such as climate regulation, fertile soil, water, and clean air (MEFCC, 2017). Forests regulate ecosystems, protect biodiversity, play an integral part in the carbon cycle, support livelihoods, and supply goods and services that can drive sustainable growth (MEFCC, 2017).

The forest sector is a strong element of the Sustainable Development Goals (SDGs), particularly SDG 13 Climate Action and SDG 15 Life on Land. A recent study (MEFCC, 2016, unpublished) reports that in 2012-13, Ethiopia's forests generated economic benefits in the form of cash and in-kind income equivalent to USD 16.7 billion (111.2 billion Ethiopian Birr (ETB)), or 12.9% of the measured value of GDP. The largest market income benefits were associated with flows of wood fuel (firewood and charcoal) and livestock fodder from forests. In addition, 2.4 billion ETB was attributed to non-market benefits based on Ethiopians' willingness to pay to maintain forests. Further, the study suggests that current GDP estimates undervalue the contribution of the forest industry to national income by about 42%. Their assessment estimates the forest industry's contribution to GDP to be about 6.1% (MEFCC, 2017).

Ethiopia formulated its Climate Resilient Green Economy strategy in 2011(CRGE, 2011). The CRGE intentionally aim at following a carbon-neutral and climate-resilient development trajectory, and outlines a roadmap to attain middle-income status by 2025 while building climate resilience and achieving green growth with a zero-net increase in greenhouse gas (GHG) emissions from 2010 levels, by 2030.

The country also submitted its Intended Nationally Determined Contribution (INDC) to the United Nations Framework Convention on Climate Change (UNFCCC) on 10 June 2015, which later converted to Ethiopia's 1st Nationally Determined Contribution after Ethiopia, ratified the Paris Agreement (PA) in March 2017. The updated emission level in 2010 are estimated at 247 million metric tons of carbon dioxide equivalents (Mt CO_2e) which are projected to increase to a level of 403.5 Mt CO_2e in the Business as Usual (BAU) scenario in 2030 (FDRE, 2021). The unconditional pathway will result in absolute emission levels of 347.3 Mt CO2e in 2030, which represents a reduction against the revised BAU of 14% (-56 Mt CO2e) in 2030. The impact of further policy interventions proposed under the conditional pathway decreases absolute emission levels to 125.8 Mt CO2e such that the combined impact of unconditional and conditional contributions represents a reduction of 68.8% (-277.7 Mt CO2e) in comparison with the revised BAU emissions in 2030.

The Ethiopian forest stock of 17.2 million hectares is threatened by an annual deforestation rate of 0.54% i.e., an annual loss of about 91,735 hectares despite an afforestation of 18,928 ha annually between the year 2000 and 2013 (UN-REDD, 2017). According to the Ethiopian Forest Reference Emission Level (FREL) report, emissions from deforestation stands at 17.9 Mt CO2e/year while removals from afforestation/reforestation are at 4.8 Mt CO2e/year highlighting significant loss of net forestland.

Several pledges have been made at various scales: national, regional, international, to implement climate change mitigation and adaptation actions. Parties to the UNFCCC have committed to limit the global temperature rise well below 2 degrees Celsius by 2030 on the PA (PA, 2015). Such ambitions and global targets require the attention and dedication of signatory parties to implement mitigation and adaptation actions, as well as to report on progress for global accounting. Ethiopia, as a signatory of the PA, is expected to provide timely report on progresses being made to achieve the NDC economy wide targets.

Reporting from the Forestry sector is essential to global accounting as forests play a key role in climate change mitigation, being vital sequesters of carbon. They can also be large emitters of carbon dioxide and other greenhouse gasses if deforestation and forest degradation are not reduced or halted. Measurement, reporting, and verification (MRV) requirements are put in place for transparent emission reduction accounting and for providing safeguards against environmental and social risks. Countries are expected to develop a well-established national forest monitoring system (NFMS) and MRV system to measure and report progress made on reducing deforestation and forest degradation (D&D) which can be later verified by independent parties.

Ethiopia, as one of the signatories of the Conference of the Parties (COP), has identified Agriculture, Forestry, and Other Land Uses (AFOLU) as a key mitigation and adaptation sector in combating climate change. Reports show that the forest cover of Ethiopia has increased from

11.1% in 2010 to 17.2% in 2020, which is relatively lower compared to the East African average, which is 18.2% (FAO, 2010; EFD, Unpublished; World bank, 2020).

Reliable estimation of forest area and changes is an important component in the MRV system for forest resources assessment and monitoring and facilitation of result-based payments. Accordingly, there are remarkable efforts in forest mapping and forest change detection made by different local and international partner institutions over different periods using remote sensing data. In this regard, one can mention as examples the national land cover map and forest cover map produced by the then Ministry of Environment, Forest and Climate Change (MEFCC) with the support of FAO for the year 2013, the national forest cover map produced by Environment Forest and Climate Change Commission (EFCCC) with the support of EU for the year 2020 and the 2016 African Regional land cover map produced by the European Space Agency (ESA). However, due to the different forest definitions adopted by these institutions while producing the maps, and variation in the reported accuracy of the respective maps, varying area estimates have been derived and communicated. For example, the Global Forest Resource Assessment (GFRA, 2020) has indicated that Ethiopia's forest assessment report (FAR) in 2020 is different from what was reported in 2015 due to the different forest map products adopted across reporting periods. Most importantly, a national map that abides with the latest definition of forests adopted by the Ethiopian Forestry Development (EFD) remains to be produced. The EFD defines forest as "land covering at least 0.5 ha of its area by trees with a height of at least 2 m and a canopy cover of at least 20% or trees with the potential to reach to these thresholds in-situ in due course" (UN-REDD, 2017).

Several maps (e.g., ESA land cover map, NASA's Forest Canopy Height map, Global Forest Change map) exist at regional and global scales. However, such platforms often provide information that is either generalized or too specific to a certain region, thus lacking the required level of accuracy and detail to support targeted monitoring and decision making at national and sub-national levels. Recent trends show that there is a growing availability of new datasets (e.g., high and moderate resolution optical and radar datasets such as Plant NICFI L1, Sentinel 2 and Sentinel 1, etc.), advanced analytical methods (e.g. machine learning algorithms), cloud based geospatial analysis platforms (e.g., Google Earth Engine, SEPAL, CEO, etc.), and analytical tools, along with the rising number of national experts on the field that can enhance existing products.

These trends can enable the design of a transparent, reproducible, and reliable forest map products. Such technical processes need to be guided through the coordinated effort of national MRV stakeholder institutions and experts in support of generating quality national reports for global communication.

Countries are expected to follow global guidelines in developing the definition of forests and establish a benchmark: forest reference level (FRL) and forest reference emission level (FREL) against which changes in emission reduction are measured. Hence, this forest mapping activity adopted the new forest definition of Ethiopia to produce a forest map for the year 2023 and a forest cover change map between 2020 and 2023. This will enable EFD to maintain reliable information on the country's status of forest cover change and drivers of deforestation and forest degradation.

1.2 Objectives

The overall aim of the mapping exercise was to undertake forest cover change detection and develop standardized forest maps consistent with the forest definition adopted by Ethiopian Forestry Development (EFD) that will guide the delineation, demarcation and certification of forestlands.

2. Methodology

2.1 Satellite data acquisition

2.1.1 Planet NICFI

In order to map forestland for the year 2023 and a forest cover change map between 2020 and 2023, a high spatial resolution Planet NICFI level 1 imagery was acquired for the years 2020 and 2023 covering the boundary of Ethiopia. Planet NICFI level 1 quads for 2020 and 2023 were downloaded and mosaicked using System for Earth Observation Data access, Processing, & Analysis for Land Monitoring (SEPAL) for the two years considering a relatively low cloud cover period of the year, specifically for the month of December – January (Figure 1).

2.1.2 Method

SEPAL is a web-based cloud-computing platform that enables users to create image composites, process images, download files, create stratified sampling designs, and more activities simply using internet browser platform (Figure 2). It is designed by the United Nation's Food and Agricultural Organization (FAO) to support the remote sensing and satellite-based forest monitoring efforts of developing countries. Planet NICFI mosaics were created from 01 December 2019 to 31 January 2020 and from 01 December 2023 to 31 January 2024 for the year 2020 and 2023, respectively (Figure 1).

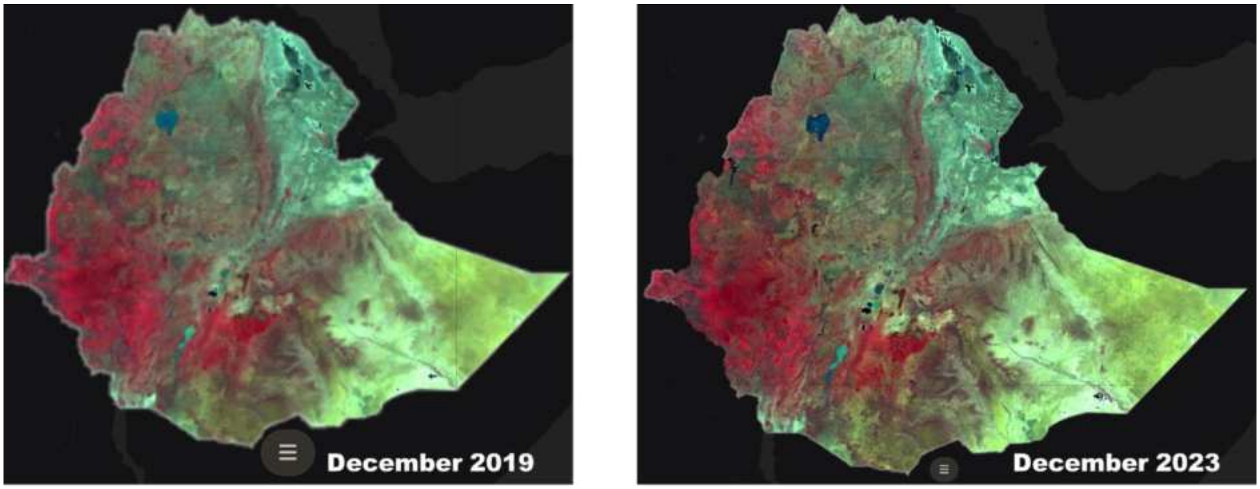

Figure 1: Planet mosaics ready for analysis for 2020 and 2023

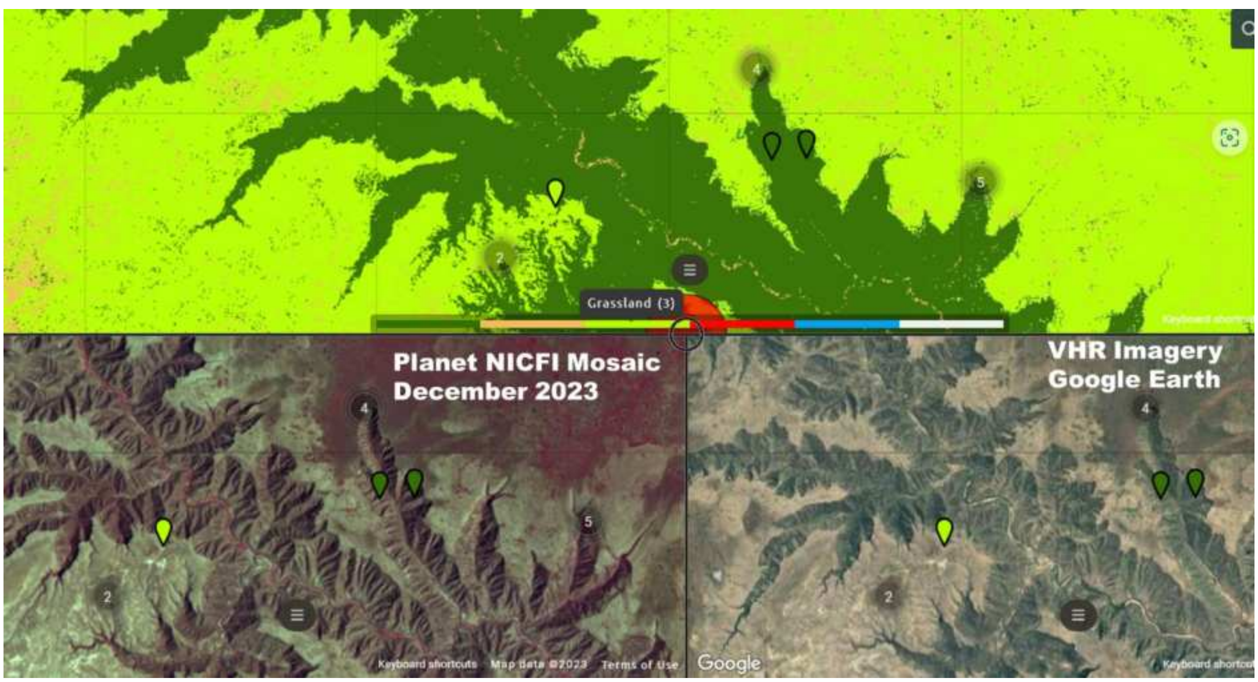

Figure 2: SEPAL interface

The forest mapping activity is performed by dividing the country into various areas of interest (AOI). The area of interest (AOI) is subdivided in to 10 classes based on the vegetation type of Friis and Demissew (2010) and agro-ecological zones (AEZ) of Ethiopia (Figure 3). Homogeneity in terms of topography (elevation) and climatic parameters such as rainfall and temperature were attained. This helped us to classify land features correctly according to the vegetation types, which ultimately increases the accuracy of the LULC maps.

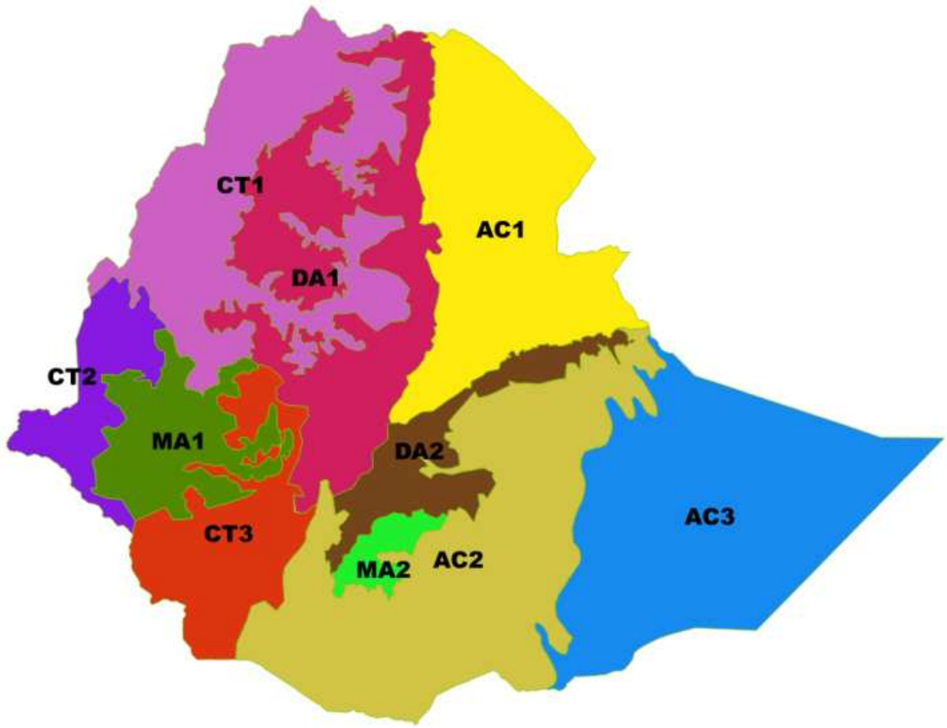

Figure 3. Subdivision of the AOI

AC = *Acacia-Commiphora*; CT = *Combretum-Terminalia*; DA = Dry Afromontane; MA = Moist Afromontane

2.2 Class definition

The LULC maps have six classes such as Forestland, Cropland, Grassland, Settlement, Wetland and Other land according to Intergovernmental Panel on Climate Change (IPCC) LULC classes. According to IPCC (Eggleston, 2006; IPCC, 2014, 2003) forestland includes all land with woody vegetation consistent with thresholds used to define forestland in the national GHG inventory, sub-divided into managed and unmanaged, and also by ecosystem type as specified in the IPCC Guidelines. It also includes systems with vegetation that currently fall below, but are expected to exceed, the threshold of the forestland category.

According to the forest reference emission level (FREL/FRL) of Ethiopia (UN-REDD, 2017) ‘forest’ is defined as 'Land spanning at least 0.5 ha covered by trees (including bamboo) (with a minimum width of 20 m) attaining a height of at least 2m and a canopy cover of at least 20% or trees with the potential to reach these thresholds in situ in due course.’ (MEFCC minute cited in UN-REDD, 2017). Non-forest can be defined as any LULC which is neither forest nor bamboo.

Deforestation/Forest loss is defined as a human-induced, permanent conversion of forest land to another land use or land cover within a spatial resolution of the satellite imagery used.

Forest gain is defined as a permanent conversion of another land use or land cover to forest land within a spatial resolution of the satellite imagery used in a given period of time which includes assisted natural regeneration.

2.3 Mapping scheme and mapping process

The mapping process includes imagery data acquisition, training data collection, pre-processing (image stacking, clipping, enhancement and mosaicking) through SEPAL, image classification through SEPAL, post-processing, accuracy assessment and generation of bias corrected area estimates using Collect Earth Online (CEO), R-Shiny App and MS Excel (Figure 6) and Random forest machine learning algorithm were for image classification.

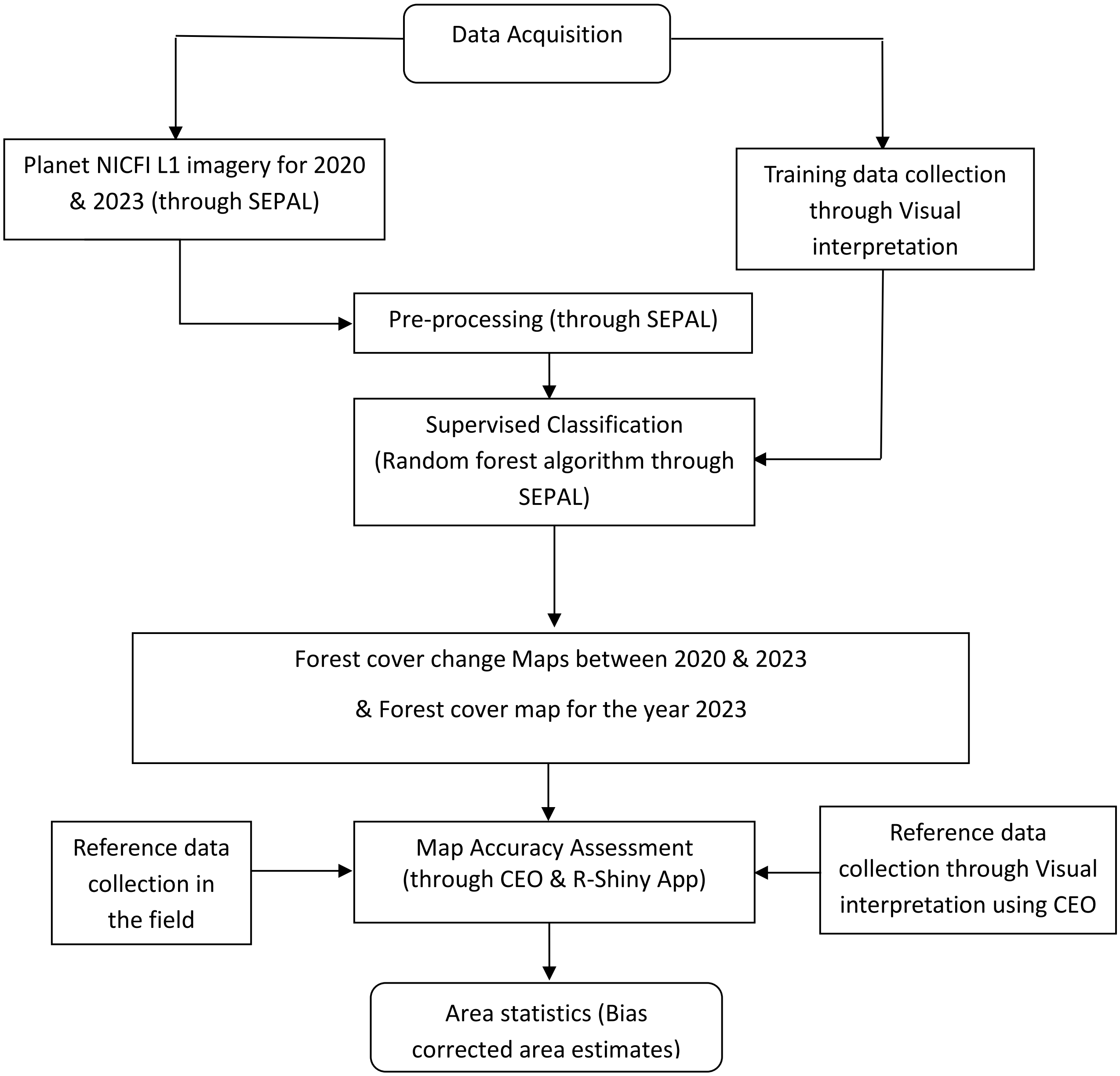

Figure 4. Workflow of forest cover mapping of Ethiopia

2.4 Training data collection

About 16,706 training data were collected for deforestation (3876) and stable (12,830) classes for the image-to-image change detection with visual interpretation of Very High Resolution (VHR) imagery from Google Earth using SEPAL covering the whole country for training the random forest classification algorithm (Figure 5). Demonstrations of training points per class are presented below in SEPAL (Figure 6-12).

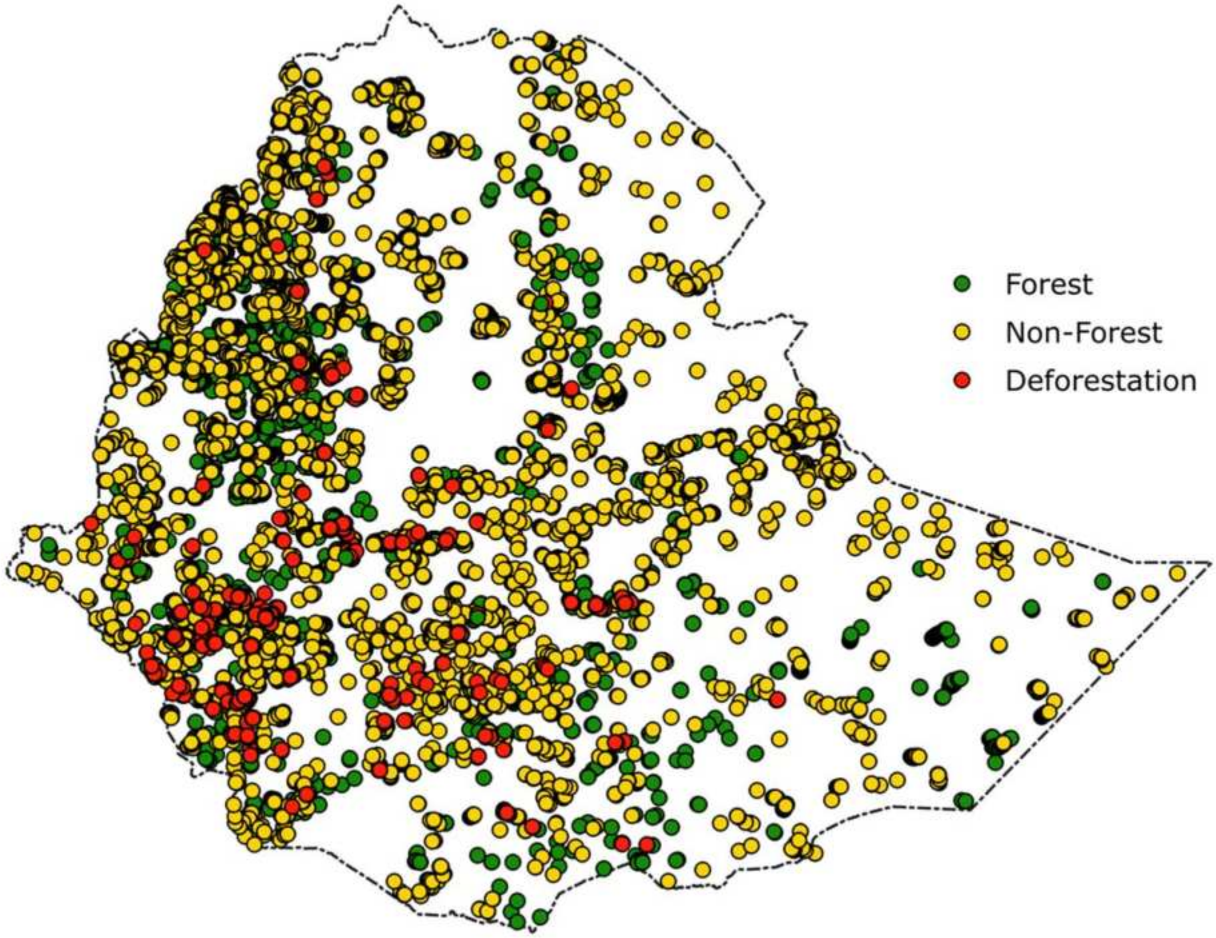

Figure 5: Training dataset for the image-to-image change detection

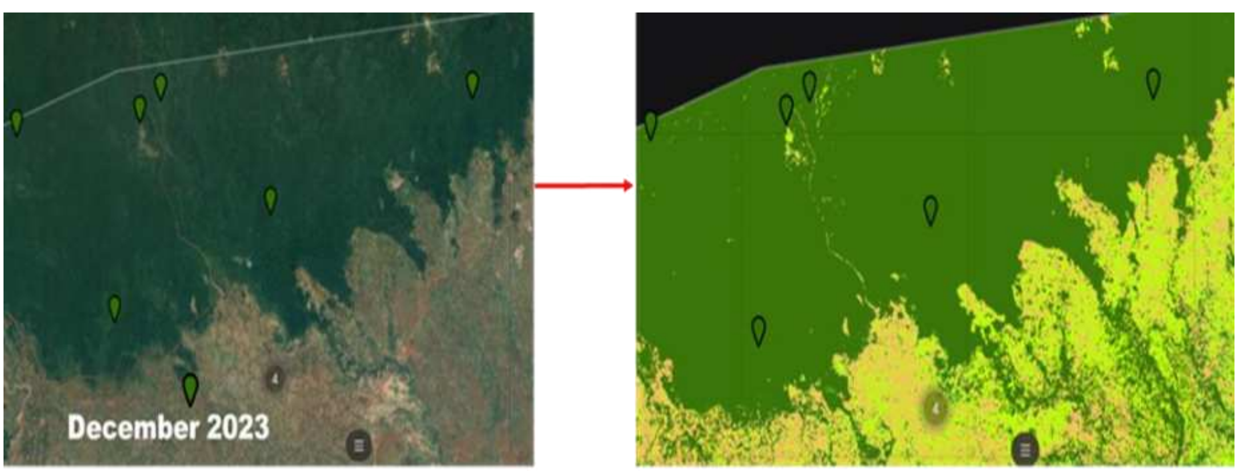

Figure 6: Training data collection for Stable Forest in SEPAL for the year 2023

Figure 7: Training data collection for Stable non-forest in SEPAL for the year 2023

Figure 8. Training data collection for deforestation (forestland converted to cropland) (Red points) in SEPAL

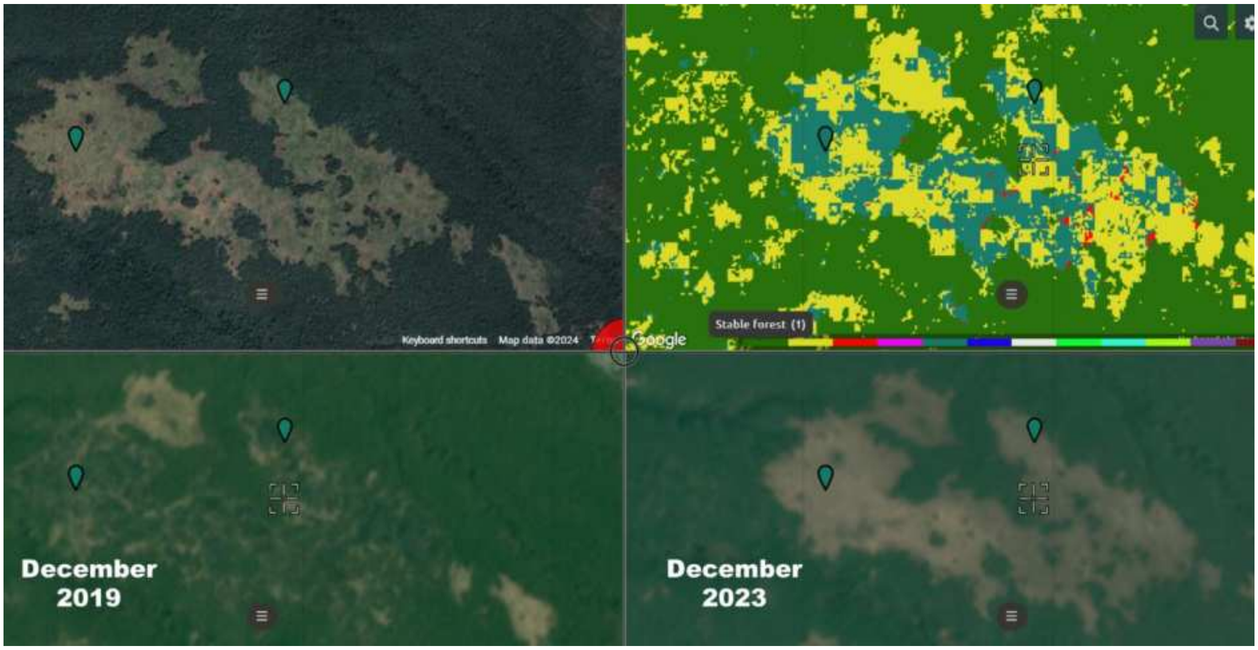

Figure 9: Training data collection for deforestation (Forestland converted to Grassland) in SEPAL for the year 2023

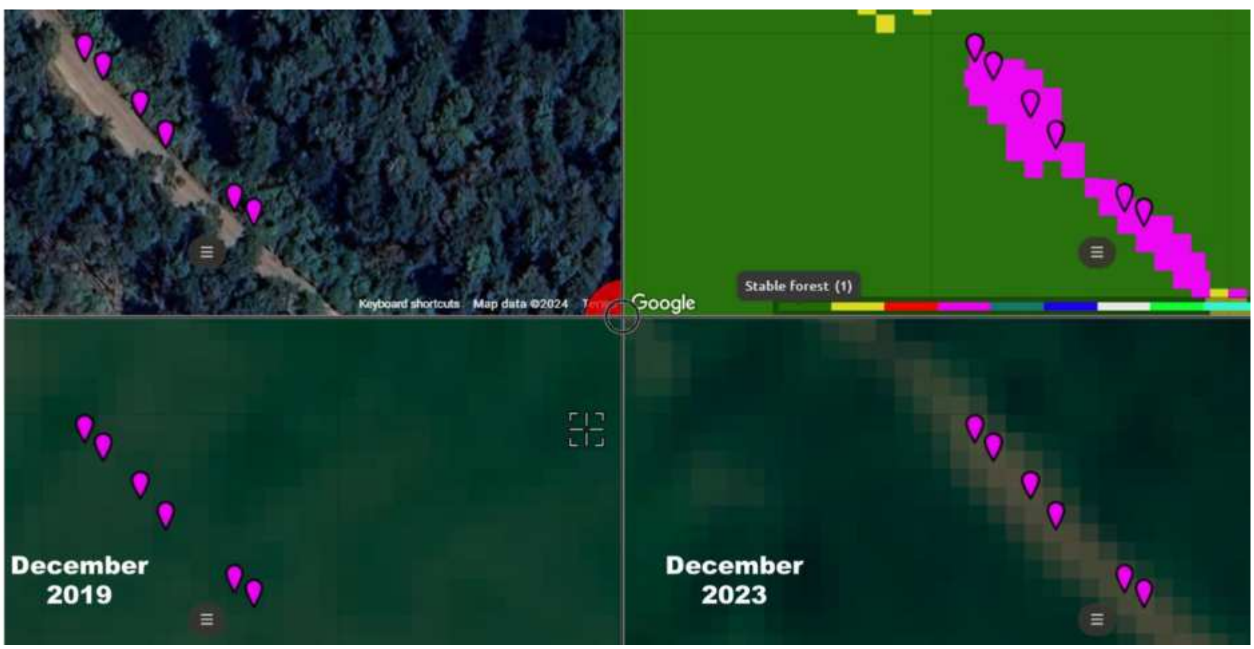

Figure 10: Training data collection for deforestation (Forestland converted to Settlement) in SEPAL for the year 2023

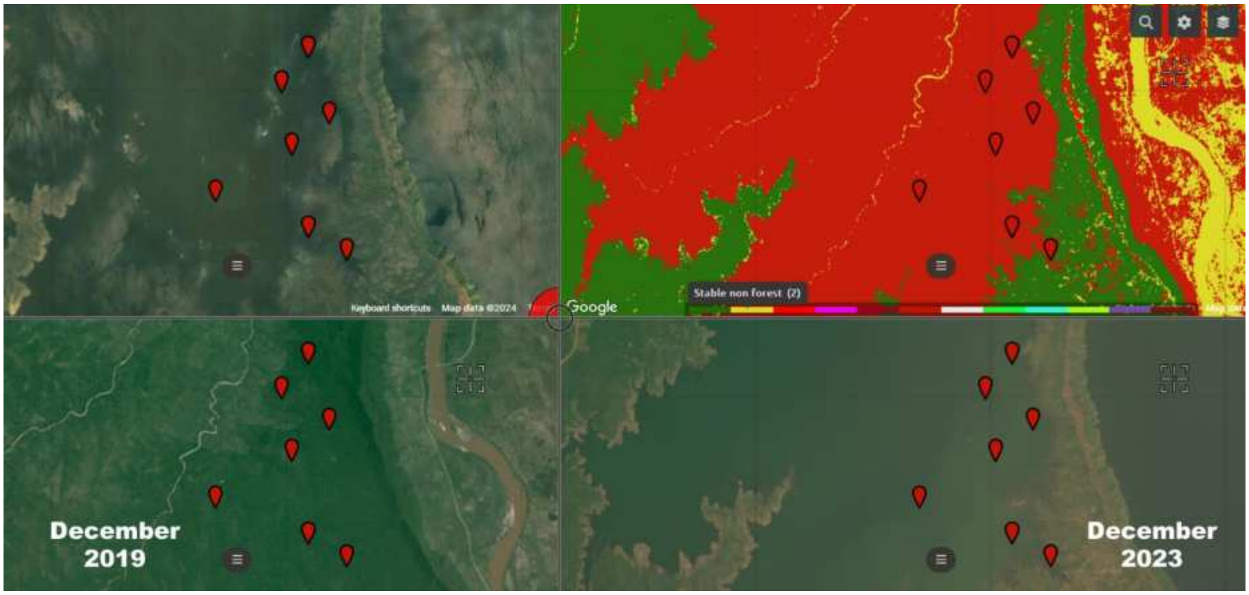

Figure 11: Training data collection for deforestation (Forestland converted to Wetland) in SEPAL for the year 2023

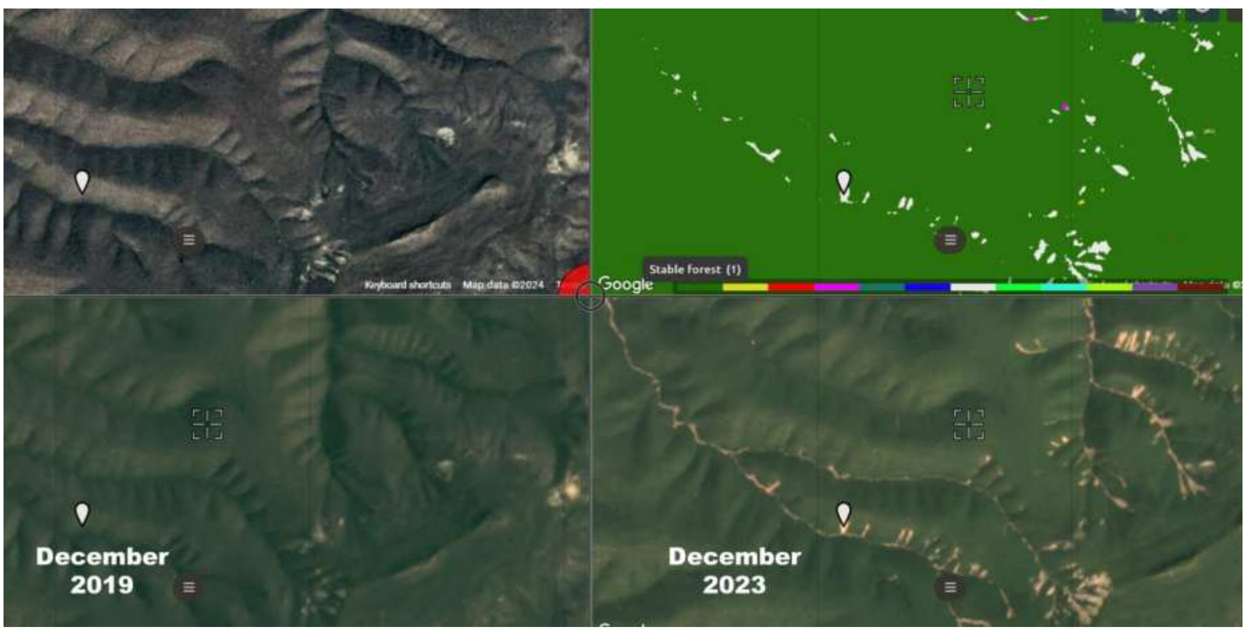

Figure 12: Training data collection for deforestation (Forestland converted to Other land) in SEPAL for the year 2023

2.5 Image Classification and post-classification editing

Post-classification change detection is not a suitable option because most historical LULC maps do not have sufficient accuracy to derive change. Particularly, post-classification is well-known to be error-prone since the errors of each single-date classification is being combined and reproduced.

Studies examining post-classification change have shown two forest/non-forest maps can be highly accurate with user's accuracies of about 95%, the user's accuracy of the deforestation class in the change map is likely to be much lower, indicating that the forest change obtained by post-classification is inaccurate (Olofsson et al., 2013).

The approach chosen to detect LULC change was a supervised change detection using change training points (Tewkesbury et al., 2015). Therefore, image-to-image change detection was applied following the GFOI guiding principle 1 for remote sensing (GFOI, 2014): ‘When mapping forest (LULC) change, it is generally more accurate to find change by comparing images where the satellite imageries from base and end year processed in the same algorithm as opposed to comparing maps estimated from images.’ In the case of change detection, the classes are deforestation or stable. The stable ones are stable forest (forestland remaining forestland) and stable non-forest. The deforestation classes are forest loss (forestland converted to cropland, forestland converted to grassland, forestland converted to settlement, forestland converted to wetland and forestland converted to other land). Such classification enables the forestry sector to have IPCC compliant activity data (AD) for national and international reporting requirements.

The process assessed two mosaics for the year 2020 and 2023, to assess the change occurred in this period. A target day is fixed in order to get the maximum vegetation cover and least cloud cover as possible. All the data collection, correction and composition are implemented within Google Earth Engine (GEE) API (Application Programming Interface) integrated with SEPAL.

As supervised classification is dependent on the quality of samples, about 16,706 training points were generated for all of the stable and deforestation classes. Points for changes were carefully assessed through a visual assessment using a time series of Sentinel-2 images and vegetation indices and very high-resolution imagery available in the Google Earth.

Spectral bands such as Near Infrared (NIR), Red, Green and Blue were utilized during classification to get better classification. Auxiliary data sources such as SRTM data, Latitude and JRC Global Surface Water Mapping Layers, which were integrated with SEPAL, are applied during classification. The classification has been significantly improved due to the utilization of SRTM (terrain) data which includes elevation, aspect and slope, particularly considering Ethiopia's rugged topography (Figure 13).

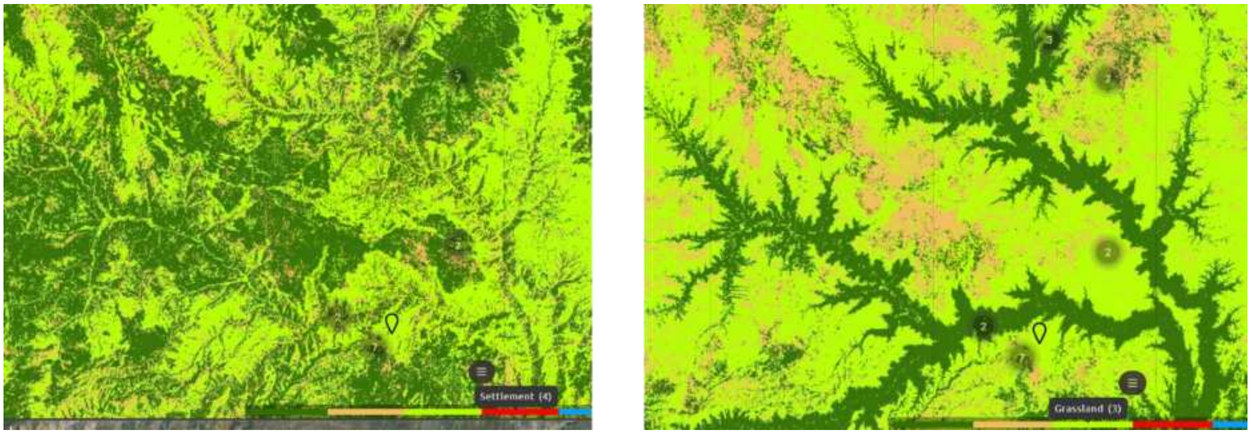

Figure 13:Classification of a riverine forest without SRTM data (Left) and with SRTM data (Right)

The selected approach for LULC mapping is supervised classification. In a supervised classification of imagery, the user identifies representative spectral samples for each of the classes in the digital image. The process assessed two mosaics for the year 2020 and 2023, to assess the change occurred in this period. A target day was fixed in order to get the maximum vegetation cover and minimum cloud cover as possible.

Despite being time-consuming and requiring extensive editing work for the misclassified pixels, the team dedicated significant effort to edit the misclassifications. The minimum area of the forest definition is 0.5 ha and hence the minimum mapping unit (MMU) of the map is 0.5 ha. Accordingly, the minimum biophysical threshold values were set for forest patch area (>=0.5 ha), tree height (>= 2m), and tree canopy cover (>=20%) for the data processing. Therefore, in order to have a 0.5 ha MMU for Planet NICFI L1 imagery of about 5 m spatial resolution 200 adjacent pixels were merged together. MMU is attained using the algorithm "sieve" in QGIS.

Post-classification manual editing using satellite images is a crucial technique in remote sensing and geospatial analysis for minimizing misclassifications. Despite the effectiveness of automated classification methods, they can still lead to misclassifications due to various factors such as spectral confusion or limitations in distinguishing subtle differences between similar LULC types. To address these issues, post-classification manual editing involves a detailed review of the classified map against high-resolution satellite imagery. This enables the analysts to identify and correct errors through visual interpretation and ensuring a higher level of accuracy. This method requires expertise as it involves the application of image interpretation elements. For instance, major towns in the southwestern part of Ethiopia were manually edited, and misclassified large wet grasslands were identified and manually digitized using Planet NICFI and Google satellite images in the QGIS environment. Additionally, coffee and tea plantations in the southwest were also manually edited features.

2.6 Map accuracy assessment

2.6.1 Reference (validation) data collection

The map's errors are identified through verification of the map classification by visual interpretation of 3599 points in CEO (Figure 15) and reference data collection on the ground (94 points) (Figure 16) which were the only ground survey data available at that time. About 3693 reference data (ground truth samples) were collected from visual interpretation using Very High Resolution (VHR) imagery from Google Earth using CEO and from the field ground truthing activities conducted in different provinces of Ethiopia for assessing the accuracy of the forest cover map for the year 2023. CEO is a tool for collecting reference data from very high, high and medium resolution satellite imageries. It was developed by Food and Agriculture Organization of the United Nations (FAO) under the Open Foris Initiative. CEO is a free and open-source image viewing and interpretation tool, suitable for projects requiring information about land cover and/or land use. CEO enables simultaneous visual interpretations of satellite imagery, providing global coverage from MapBox and Bing Maps, a variety of satellite data sources from Google Earth Engine (Figure 14).

Figure 14: CEO interface

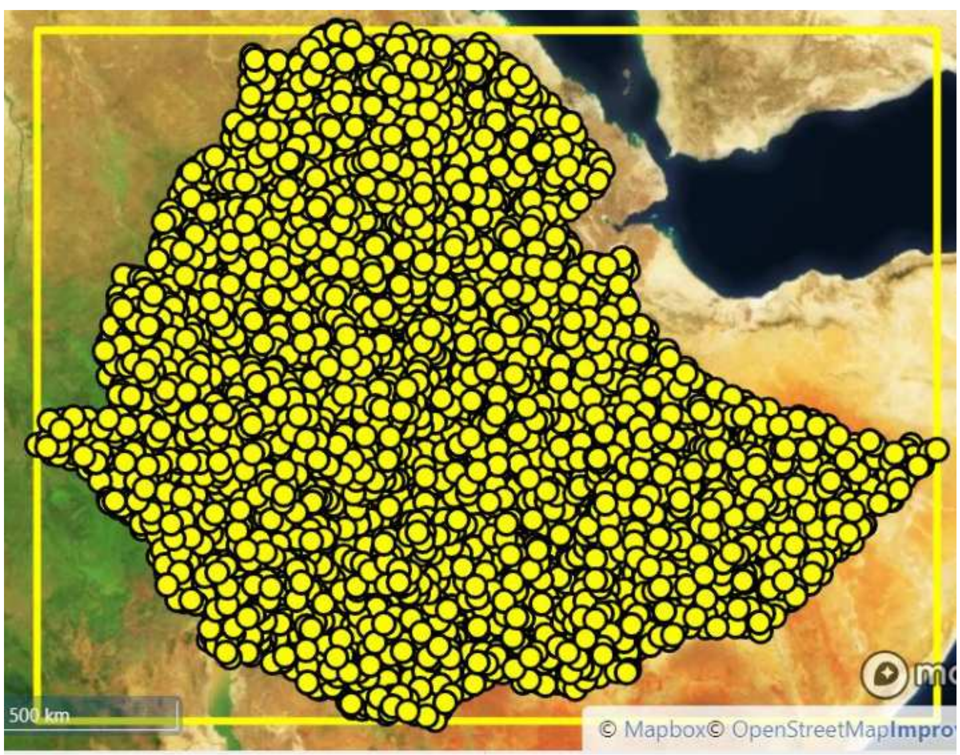

Figure 15: Reference points in CEO

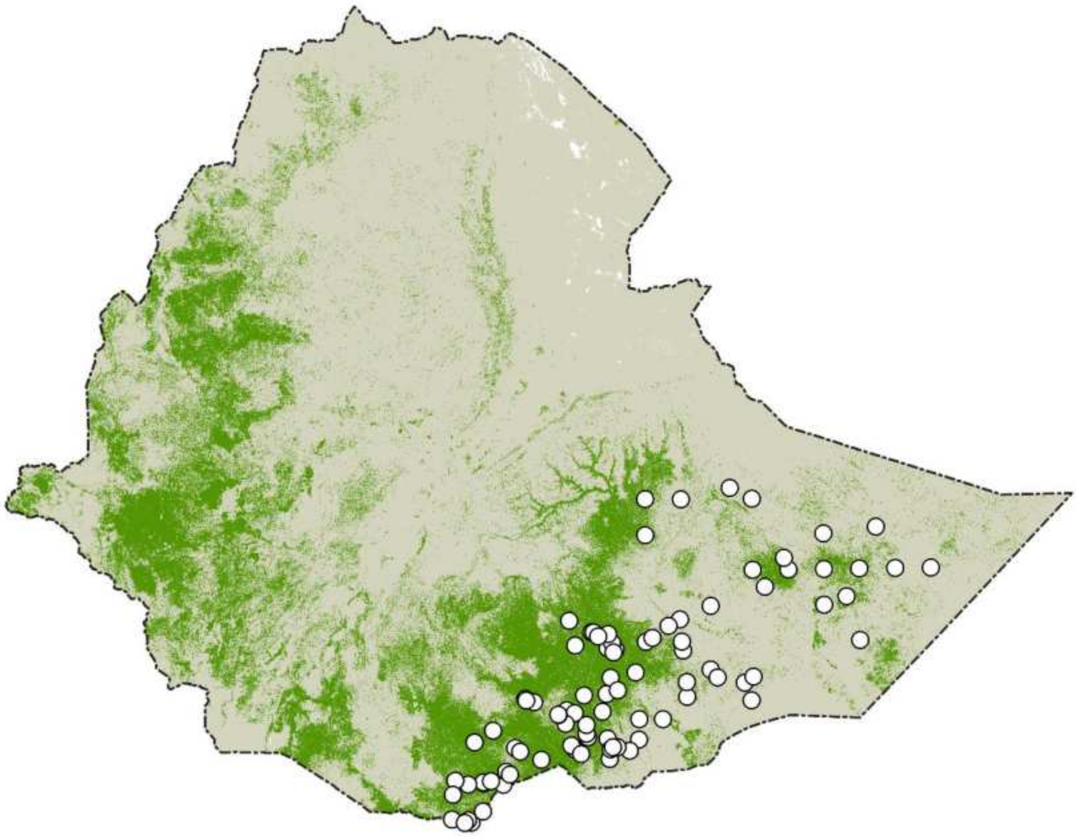

Figure 16: Reference points collected on the ground

The classified maps may contain some sort of systematic errors related with classification technique, nature of satellite data and methods of data capture, among others. According to good practice guideline (GPG) of the IPCC (IPCC, 2003), emission estimates shall neither over-nor underestimate the actual emissions as far as can be judged, and reduce uncertainties as far as practicable given national circumstances. It is also good practice to quantify uncertainties and report them in a transparent manner. A correct identification and quantification of the various sources of uncertainty helps to assess the robustness of any GHG inventory and prioritize efforts for their further development/improvement.

In order to use the classified maps, the errors were quantitatively evaluated by applying standard accuracy assessment techniques. For this study, the accuracy assessment was performed using R-shiny (for sample size calculation and accuracy estimation) and CEO (for reference data collection). R-shiny is a package in R statistical software and used to generate sample size for reference data collection and to estimate the accuracy of the maps. The reference points were generated using R-Shiny app. It has two components, accuracy assessment design and accuracy assessment analysis (Figure 17).

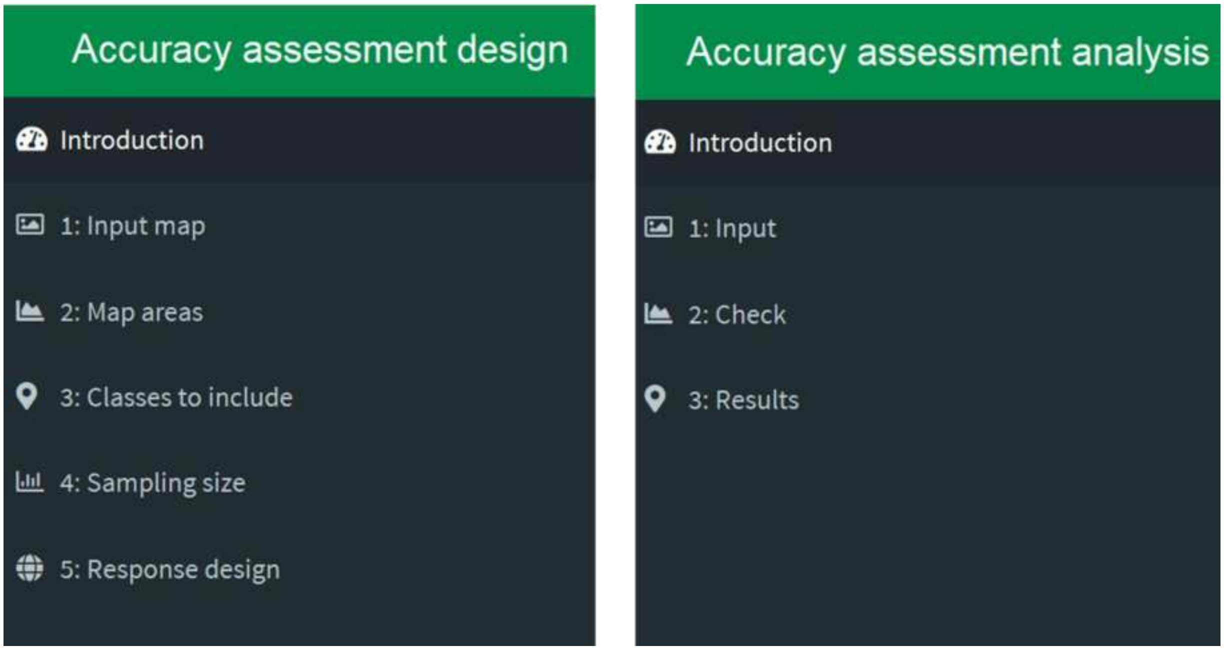

Figure 17: R-shiny app interface for accuracy assessment design and analysis

In proportional allocation, the overall sample size has been calculated to the classes proportional to the area of the classes, so rare classes have been received a small proportion of the overall sample size. The R-shiny extension used Equation 1 (Cochran, 1977) to calculate an adequate overall sample size for stratified random sampling that can then be distributed among the different classes. N is number of overall pixels, S(Ô) is the standard error of the estimated overall accuracy that will be achieved, Wi is the mapped proportion of area of class i, and Si is the standard deviation of class i.

$$\boldsymbol{n} = \frac{(\sum \boldsymbol{WiSi})^2}{[\boldsymbol{S}(\hat{\mathrm{O}})\ ]^2 + (\frac{\mathbf{1}}{\boldsymbol{N}}) \sum \boldsymbol{WiSi}^2} \approx \left(\frac{\sum \boldsymbol{WiSi}}{\boldsymbol{S}(\hat{\mathrm{O}})}\right)^2 \qquad (\mathbf{1})$$

Based on this equation, the total numbers of reference sample plots (n) used for accuracy assessment were 3693. The plot covered 0.50 ha (71mx71m).

The required overall number of samples which was calculated using the Olofsson et al. (2013) methodology adopting the following assumptions, using conservative user inputs to have a cost effective and robust sample size (Table 1):

Table 1: Inputs needed for the statistical formula that calculates the overall number of samples for the accuracy assessment.

	Input values selected
The standard error of the estimated overall accuracy that will be achieved	0.005
Expected user's accuracy for forest	0.9
Expected user's accuracy for non-forest	0.9

The total required samples for the accuracy assessment are distributed proportionally across the two map classes (Forest and Non- Forest) (Table 2). We collected reference sample data to verify the accuracy of the map classification at each sample point location. The sample data was collected through visual interpretation using a time series of very high-resolution imagery and high-resolution Planet NICFI images. The visual interpretation of sample points is referred to as "reference data". Points where the interpreters had low confidence in the classification were subject to field verification (83 points). Consequently, the total number of reference points used in the accuracy assessment and bias corrected area estimation is 3693 samples. Following the generation of reference sample plots, these reference sample plots were assessed using CEO. This tool enables users to visually assess the LULC of sample locations with the freely available data from Google Earth and high-spatial-resolution Planet NICFI images.

Table 2: Samples distributed in the map classes for verification (collection of reference data)

Classes	Sample points per map class
Forest	802
Non-forest	2797
Total samples	3599

2.6.2 Bias corrected area and uncertainty estimates

According to IPCC (Espejo et al., 2020; IPCC, 2003), it is good practice for countries to produce emission estimates which: 1) neither over-nor underestimate actual emissions as far as can be judged, introducing a systematic error (or bias), and 2) reduce uncertainties as far as practicable given national circumstances. It is also good practice to quantify uncertainties and report them in a transparent manner. All surface estimations contain errors, most of which tend to be systematic (bias).

The map area was corrected for bias using stratified random samples using visually interpreted high temporal and spatial resolution satellite imagery. Classification errors were identified by collecting sample point data, independent of the training data. The sample data verifies whether the classification is correct or incorrect at the location of the sample points. This information is summarized in an error matrix and the matrix is used to correct the areas per class for the map bias resulting in new area estimates, referred to as bias-corrected area estimates. Bias corrected areas of each class were estimated after accuracy assessment using a confusion matrix (error matrix) (Table 7). The map area estimates are corrected for systematic error (bias) in the map and confidence intervals are calculated around the bias-corrected area estimates as a measure of the random error remaining in the estimate after removal of the bias. The bias-corrected area estimates are obtained by the map area minus the over-detections (commission errors) plus the under-detections (omission errors). E.g., the bias-corrected area estimate of forest class is calculated by 28,324,359 ha - 4,449,548 ha + 2,880,125 = 26,754,937 ha (Table 7).

To quantify and report uncertainty, confidence intervals were calculated and reported around the bias-corrected area estimates, which provide an indication of the precision of the estimate. Precision gives a description of random errors or variability. Large confidence intervals indicate a large statistical variability of the population. The process of correcting the map for bias is referred to as accuracy assessment. The 90% confidence interval for the bias-corrected area estimates are calculated by multiplying 1.645006 by the standard error of the area estimate.

2.7 Quality Assurance and Quality Control (QA & QC) activities

Both formal and informal QA/QC methods were employed. Informal QA/QC involved qualitative approaches to identify issues within our own analysis and classifications enabled to iterate and create improved classifications. The team members dedicated significant time to reviewing their own classification in order to ascertain the robustness of results. This process enabled the team to visualize the data and collect additional training points for areas of poor classification.

3. Results and discussion

3.1 The forest cover change map for the period 2020-2023

A deforestation map was produced which demonstrates spatially explicit changes (Table 4, Figure 18).

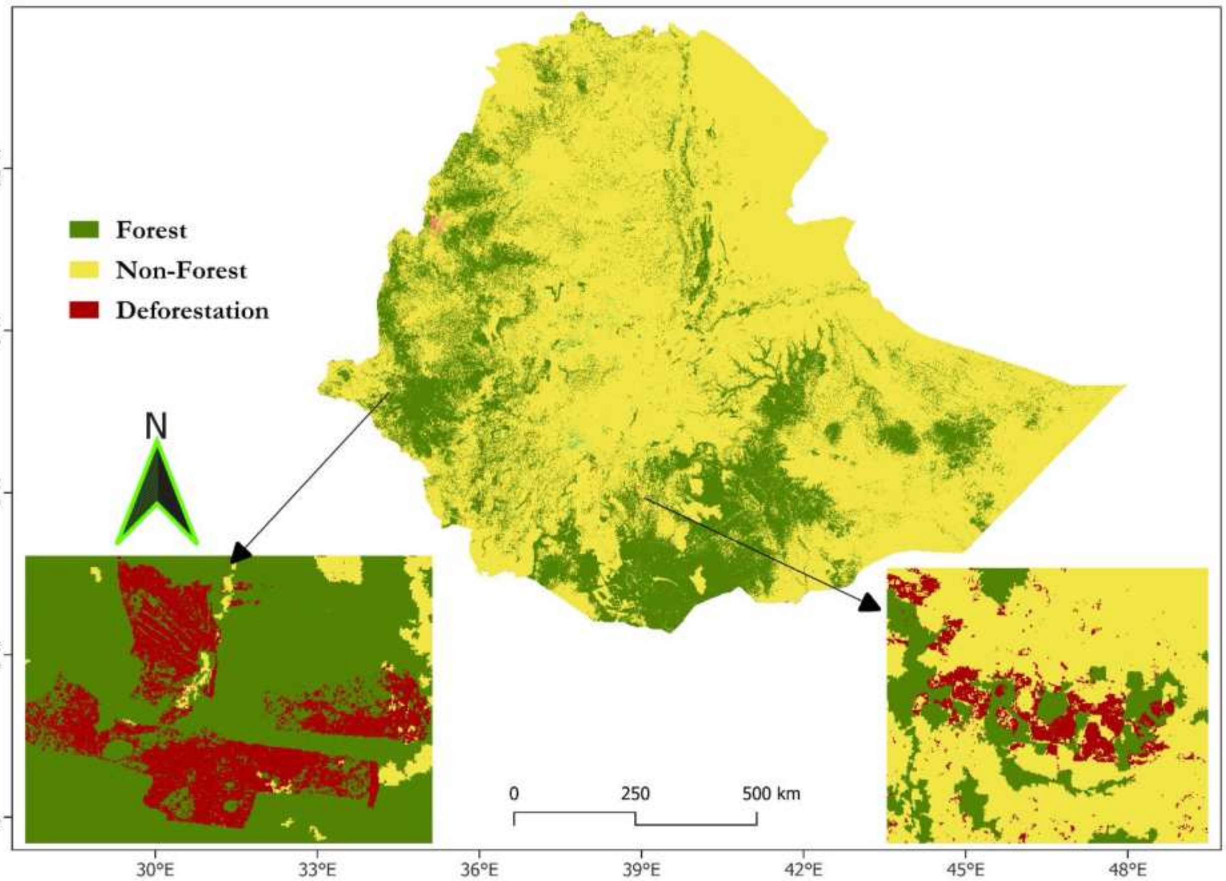

Figure 18: LULC change map of Ethiopia (2020-2023) with two zoomed in areas of deforestation

This indicates that expansion of cropland remains to be the main driver of deforestation. The sources/drivers of forest gain may include artificial and natural regeneration and expansion of small-scale tree plantation. Whereas, the sources/drivers of forest loss are such as urbanization, commercial agriculture, expansion of smallholder agricultural and commodity-based extractions. About 53 % of the deforestation is attributed to cropland followed by wetland (39 %) (Table 3).

Table 3: LULC after deforestation between 2020 and 2023

No	LULC after deforestation	AD (ha)	%
1	Cropland	58,966	53.2
2	Grassland	1,266	1.1
3	Settlement	6,256	5.6
4	Wetland	43,510	39.3
5	Other land	816	0.7
	Total	110,814	

3.2 The forest cover map for the year 2023

The three change and stable classes were aggregated into two broad categories of Forest and Non-Forest by keeping stable forest as a class of '**Forest'** and aggregating stable non forest, forestland converted to cropland, forestland converted to grassland, forestland converted to settlement, forestland converted to wetland and forestland converted to other land into a class of '**Non- Forest'** (Figure 19). The overall accuracy of the Forest / Non- Forest map for the year 2023 is 94 % (Table 4).

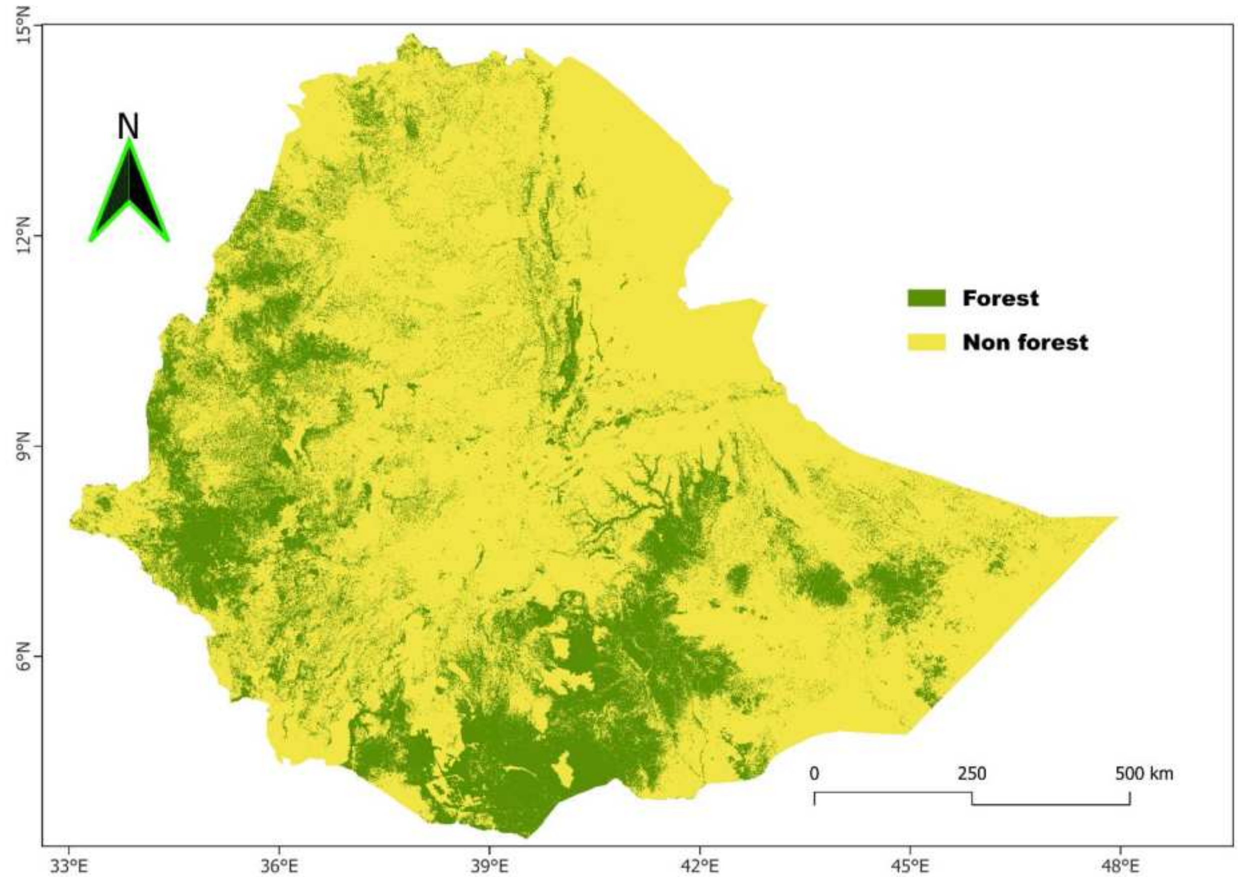

Figure 19: Forest cover map of Ethiopia (2023)

3.3 Bias corrected area estimates.

The team collected, analyzed and interpreted images for the years 2020 and 2023. Accordingly, the following statistical report for remaining land and change classes was prepared. As shown in the confusion matrix table (Table 4), the overall accuracy of the assessment is 94% with varying level of producer and user accuracy. According to IPCC GPG document assessing the accuracy of a map and using the results of the accuracy assessment to adjust the area estimates is a good practice (Hiraishi et al., 2014). Therefore, the bias corrected area estimates are summarized in Table 5.

Table 4. Confusion matrix for the forest cover map of Ethiopia for the year 2023

		Reference data			
		Forest	Non-forest	Total	**UA**
Map data	Forest	719	134	**853**	84%
	Non-forest	96	2744	**2840**	97%
	Total	**815**	**2878**	**3693**	
	PA	88%	95%		
Overall Accuracy					**94%**

PA = Producer's Accuracy (Omission errors); UA = User's Accuracy (Commission errors)

Table 5: Map and adjusted (bias corrected) area estimates with uncertainty estimates for the forest cover map of Ethiopia for the year 2023

Classes	Map area (ha)	Commission errors (ha)	Omission errors (ha)	Adjusted area estimate (ha) = map area - Commission errors + Omission errors	90% CI (ha)
Forest	28,324,359	4,449,548	2,880,125	26,754,937	750,602
Non-forest	85,203,704	2,880,125	4,449,548	86,773,126	750,602
Total	113,528,063	7,329,673	7,329,673	113,528,063	

The uncertainty estimate showed that the non-forest class is more certain than the forest class (Table 5, Figure 20). The forest cover by the year 2023 is 23.6 %.

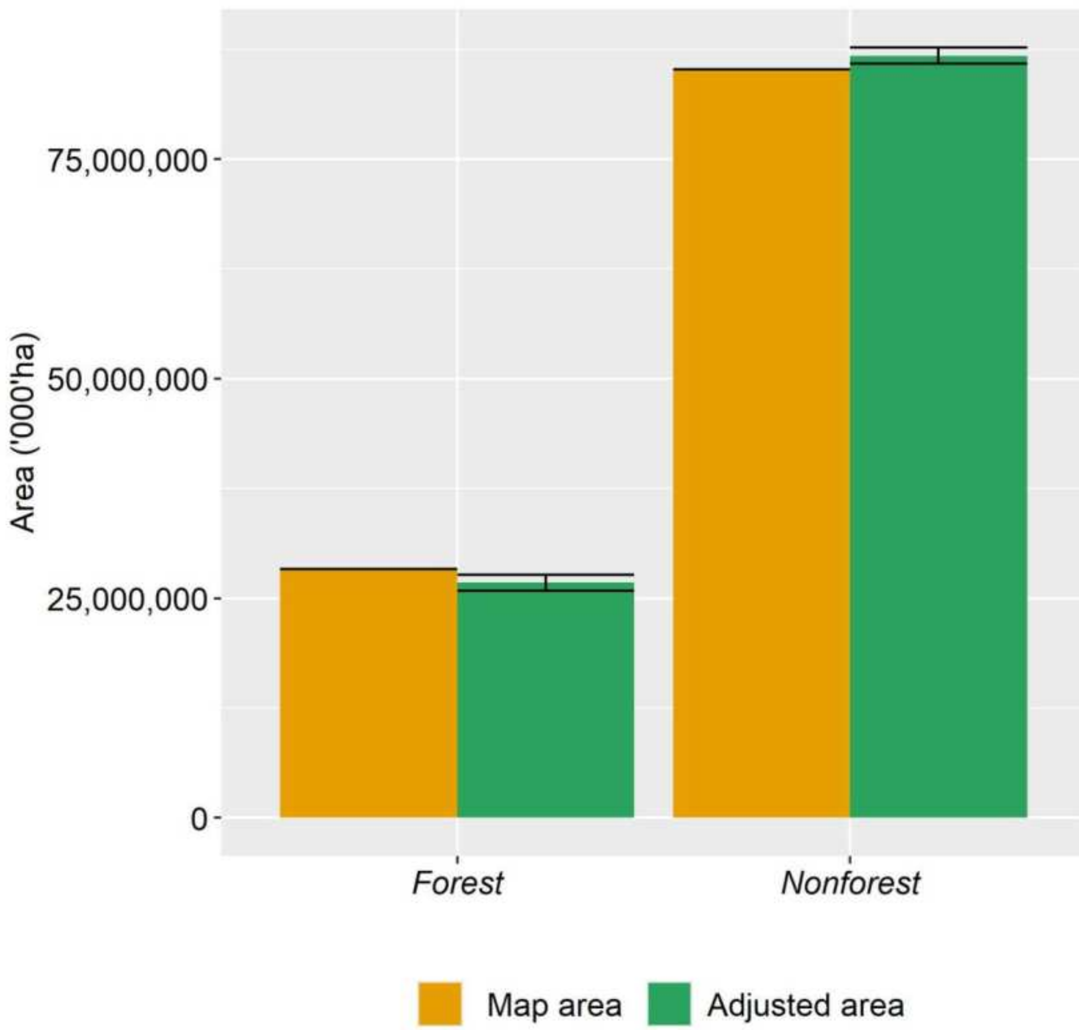

Figure 20: Map area and BCA estimates with uncertainty estimates for the final forest cover map of Ethiopia for the year 2023

3.6 Activity data trend analysis

Having spatially explicit LULC change detection (Figure 18) leads to Approach 3 (The last approach) of AD generation according to IPCC GPG document. Deforestation was declining from the previous periods onwards to the current period (Figure 21, Table 6). This is probably linked to the decline in large scale commercial agricultural investment at the expense of natural forests (changing investment policy) and boosting Participatory Forest Management (PFM) in south-western Ethiopia. PFM acknowledges community participation in forest management to safeguard forests while allowing their traditional forest use. Moreover, the Green Legacy Initiative (GLI) by itself created awareness about conservation of natural forests. Cultural values, traditional knowledge and institutions also contributing a lot for forest conservation. On the other hand, forest gain shows an increasing trend when the current period was compared to the previous periods (2000 – 2013 (UN-REDD, 2017) and 2014 – 2020 (EFCCC, 2020)) (Table 6).

Table 6. Activity Data (AD) trend per period

Class	2000 - 2013	2014 - 2020	2020 - 2023
Forest Loss (ha)	1,192,559	229,163	110,814
Forest Loss (ha/yr)	91,735	38,194	27,703
Forest area at the end of the period (ha)	17,785,035	19,372,865	26,754,937
Forest cover at the end of the period (%)	15.5	17.2	23.6

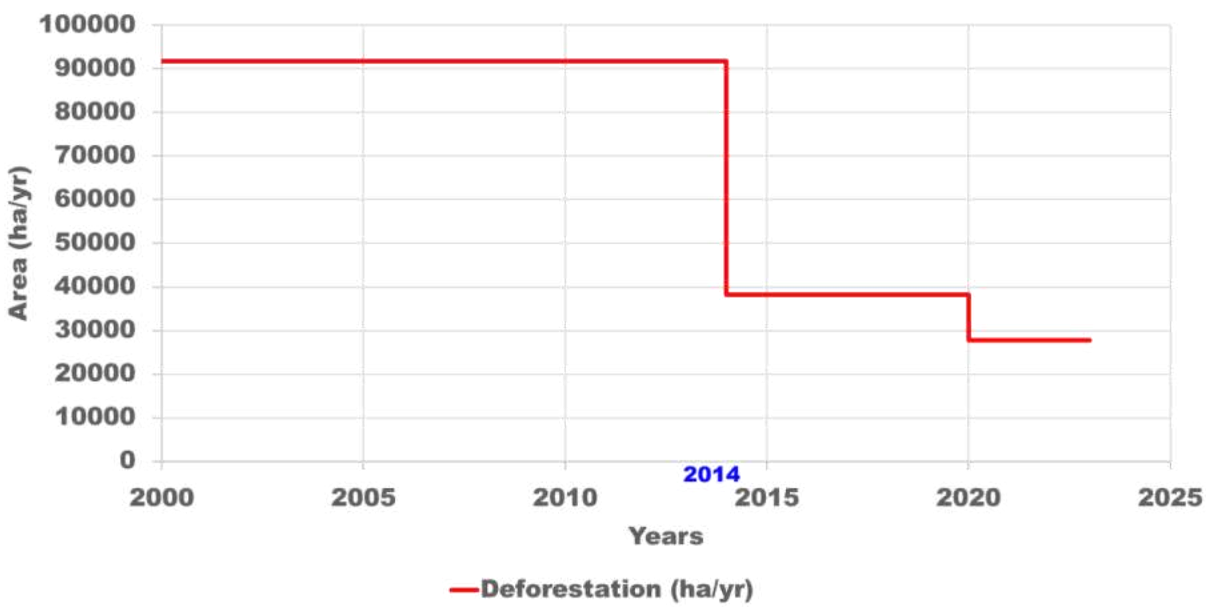

Figure 21. Deforestation trend per period

Despite the promising declining trend in forest cover loss (Figure 21), it is worth mentioning cropland expansion and commodity driven deforestation would continue to occur and reverse the magnitude unless permanence of the observed decline is maintained through strong commitment and policy support coupled with accelerated investment in forest sector mitigation actions.

4. Conclusions and Recommendations

4.1 Conclusion

CIFOR in collaboration with CIAT and EFD conducted the national forest mapping. High resolution imagery (Planet NICFI Level 1) is used for the first time for the national scale mapping in Ethiopia. The team has also applied a contemporary state-of-the-art techniques and tools (SEPAL, CEO, GEE, R-Shiny App and others) for the analysis. Two distinct maps were produced including, forest cover change map for the period 2020 – 2023 and forest cover map for the year 2023. The overall accuracy of the forest cover map for the year 2023 is 94 %. The forest cover by the year 2023 is 26,754,937 ha (23.6 %). This product represents a significant advancement compared to previous national, regional, and global mapping products for Ethiopia, introducing advanced methodologies and tools and improved classification accuracies. Deforestation is decreasing compared to the historical reference period (2000 – 2013).

4.2 Recommendations

The current forest mapping activity utilized state of the art models, processing algorithms and high-resolution multispectral satellite imagery. This significantly improved the accuracy of the forest mapping activity. Therefore, the following are some recommendations:

- Strengthening collaboration with regional and local institutions (academia, research institutions) for quality assurance and state-of-the art research.
- Transforming towards use of high-resolution satellite data for land monitoring is started. It is vital to strengthen the experience and improve the capacity of national experts for continuous use of high-resolution data and for future forest mapping activities.
- Promote and disseminate the generated data and information for wider stakeholders.
- Publishing online of the results soon for assuring open and transparent data/ Data democratization
- Going to Results Based Payments (RBPs).

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